

A380X computed ALTN fuel and time

The FCOM default computation of the fuel and time to the alternate, instead of a fixed 6.5 t

Branch	pr/20-a380x-altn-fuel-computation
Aircraft	A380X (MFD FUEL&LOAD and ALTERNATE pages)
Commits	3c3ef82b7 feat(a380x/mfd): computed ALTN fuel and time (FCOM default computation)
Base	pr/10-a380x-mfd-fcom-pages (FUEL&LOAD page)
Status	CI checks pass.

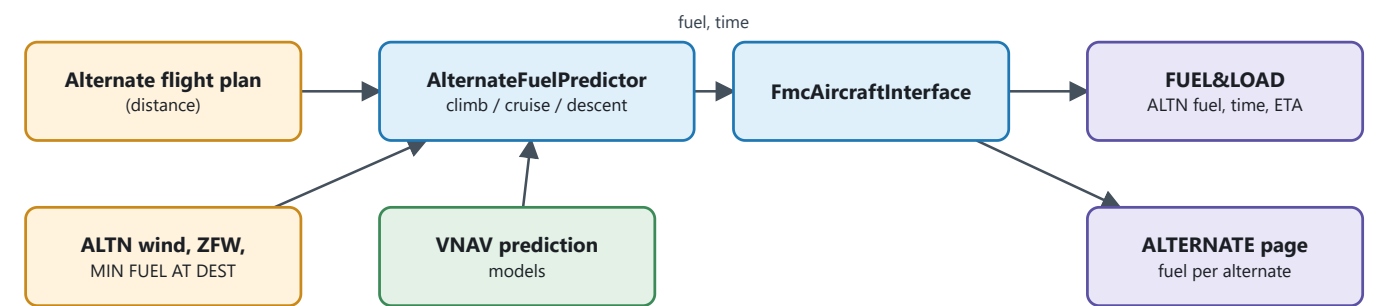
Purpose

The ALTN fuel of the FUEL&LOAD page was a fixed 6.5 t, whatever the alternate.

Specification (A380 FCOM DSC-22-FMS-20-30, FUEL&LOAD page)

- ALTN fuel (time): the trip from the primary destination to the landing at the alternate, computed with CI 0.
- Cruise at FL220 when the alternate flight plan distance is less than 200 NM, FL310 otherwise; lowered when the distance is too short for the climb and the descent.
- Climb at climb thrust, cruise, idle descent, with the VNAV prediction models of the aircraft at the FMS CI 0 speeds (290 kt, 288 kt in descent, M0.84, 250 kt below FL100).
- Distance: the alternate flight plan without its missed approach; the direct distance without an alternate route. Wind: the ALTN wind. Weight: at destination (ZFW + MIN FUEL AT DEST).
- No ALTN fuel without a ZFW; recomputed only when its inputs change.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Displays

- FUEL&LOAD: the ALTN time next to the ALTN fuel, and the arrival at the alternate (destination + ALTN time).
- ALTERNATE page: the fuel to each alternate, with the same computation.

Tests

5 unit tests: `fbw-a380x/src/systems/instruments/src/MFD/FMC/AlternateFuelPredictor.spec.ts`.

PR description (from the PR draft)

From `pr-drafts/20-a380x-altn-fuel-computation.md`, section "Summary of Changes" onwards.

Summary of Changes

The ALTN fuel of the FUEL&LOAD page was a fixed 6.5 t. The A380 FCOM (DSC-22-FMS-20-30, FUEL&LOAD page) defines it: the fuel (time) for the trip between the primary destination and the landing at the alternate, computed with CI 0, at FL220 when the alternate flight plan distance is less than 200 NM, FL310 otherwise.

- **AlternateFuelPredictor:** climb at climb thrust, cruise and idle descent with the aircraft's VNAV prediction models, at the FMS CI 0 speeds (290 kt, 288 kt in descent, M0.84, 250 kt below FL100). The cruise level is lowered when the distance is too short for the climb and the descent.
- **FMS:** over the alternate flight plan distance (without its missed approach; the direct distance when there is no alternate route), with the ALTN wind and the weight at destination (ZFW + MIN FUEL AT DEST). No default ALTN without a ZFW; recomputed only when its inputs change.
- **FUEL&LOAD:** the ALTN time next to the ALTN fuel, and the arrival at the alternate (destination + ALTN time).
- **ALTERNATE page:** the fuel to each alternate, with the same computation.
- 5 unit tests (`AlternateFuelPredictor.spec.ts`).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: FUEL&LOAD with the computed ALTN fuel and time and the alternate ETA; the ALTERNATE page with the fuel of each alternate.

References

- A380 FCOM DSC-22-FMS-20-30 (FUEL&LOAD page: ALTN fuel and time, default computation at CI 0, FL220 below 200 NM, FL310 above).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Flight plan with an alternate, ZFW entered: FUEL&LOAD shows a computed ALTN fuel and time (no longer 6.5 t) and the ALTN ETA.
2. An alternate closer than 200 NM against one further away: the lower fuel; the ALTERNATE page lists the fuel of each alternate.
3. No ZFW: no ALTN fuel.
4. `npx vitest run fbw-a380x/src/systems/instruments/src/MFD/FMC/AlternateFuelPredictor.spec.ts` (5 tests).